INTRODUCTION TO MACHINE LEARNING

in Python LANGUAGE

Chapter 2.2: K-Nearest Neighbors

K-Nearest Neighbors

Outline

1. K-Nearest Neighbors: Definition
2. The KNN algorithm
3. Examples of Classification using KNN
4. Example of Regression using KNN

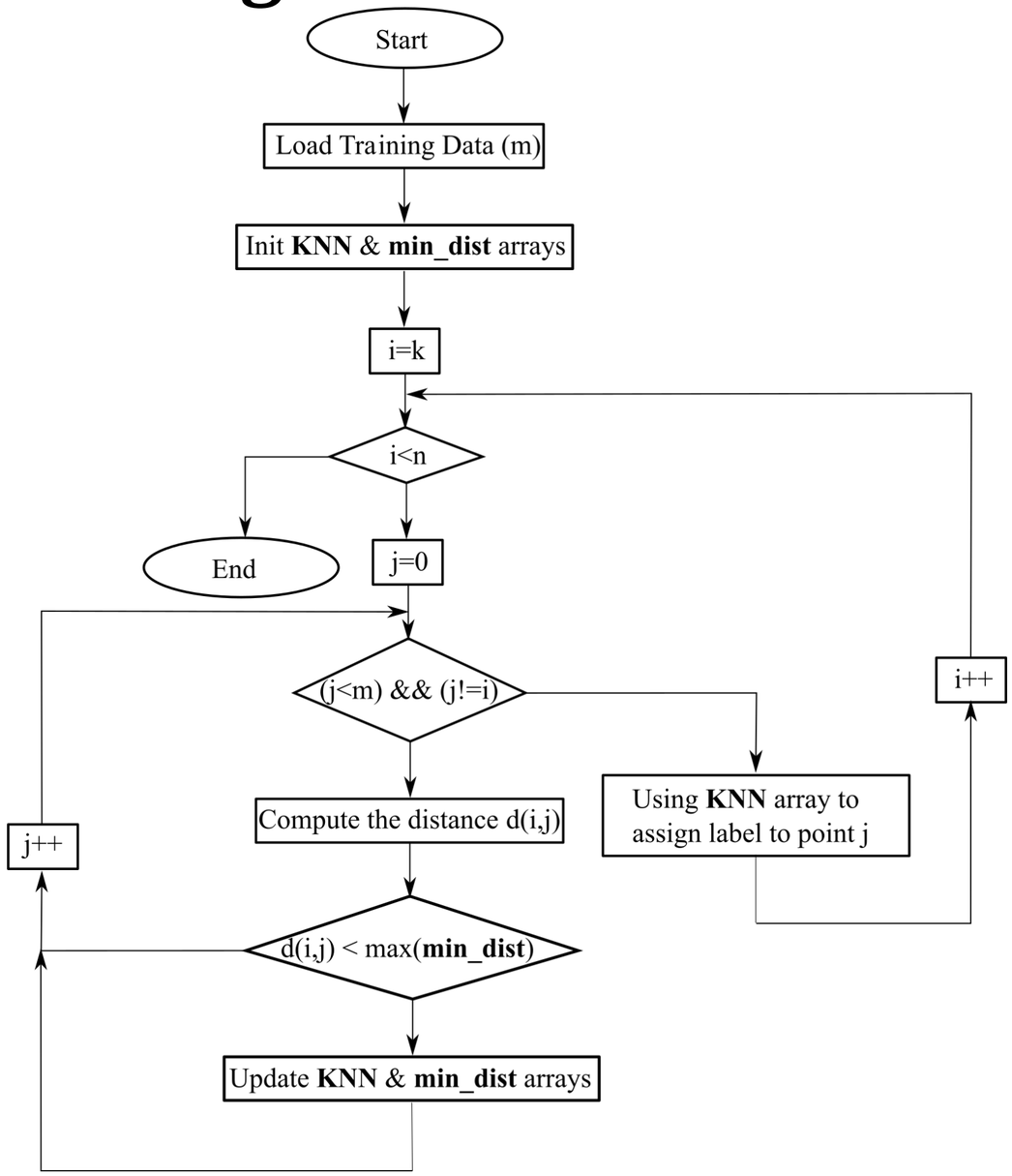
K-Nearest Neighbors

1. K – Nearest Neighbor: Definition

- KNN (K-Nearest Neighbors) is a simple and widely used machine learning algorithm for both classification and regression. The main idea is that it makes predictions by examining the 'K' closest data points in the feature space.
- KNN doesn’t build a model or learn from data, it just **stores the training data**.
- In the prediction phase, KNN computes the **distance** between the **new data point** and **all points in the training set**, typically using **Euclidean distance**. It then selects the **'K' nearest neighbors** and makes the prediction based on these neighbor points.
- Selecting the appropriate value for K (the number of neighbors) is essential. A small K can be too sensitive to noise, while a large K may generalize the prediction too much, missing important details.

K-Nearest Neighbors

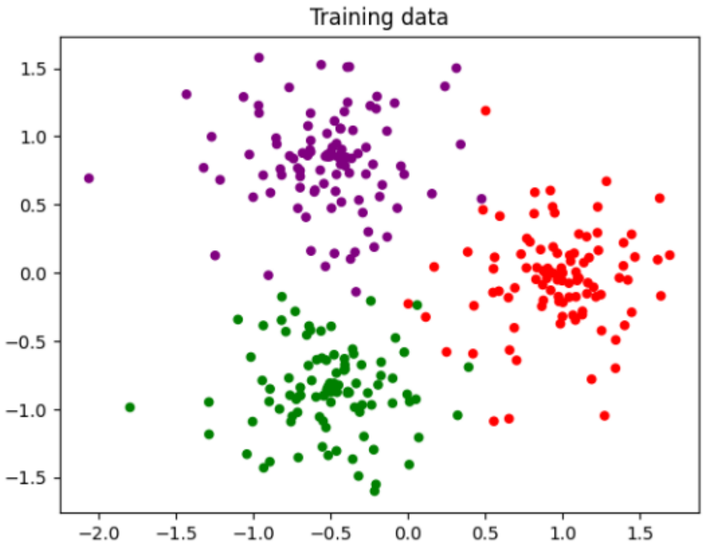
2. K – Nearest Neighbor algorithm:

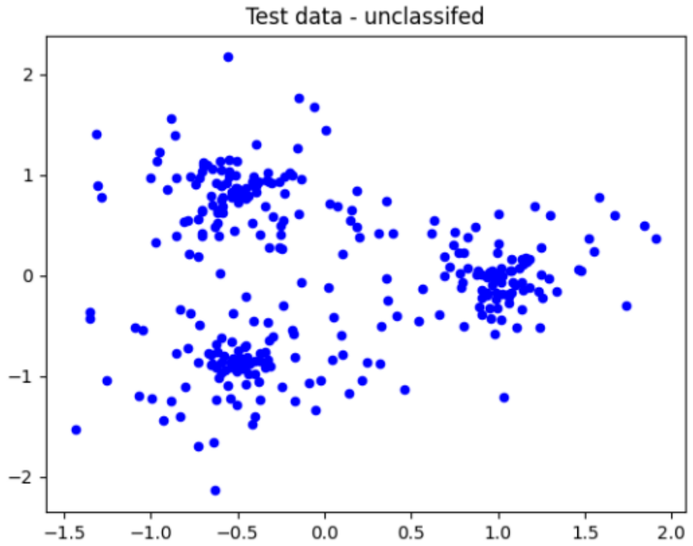


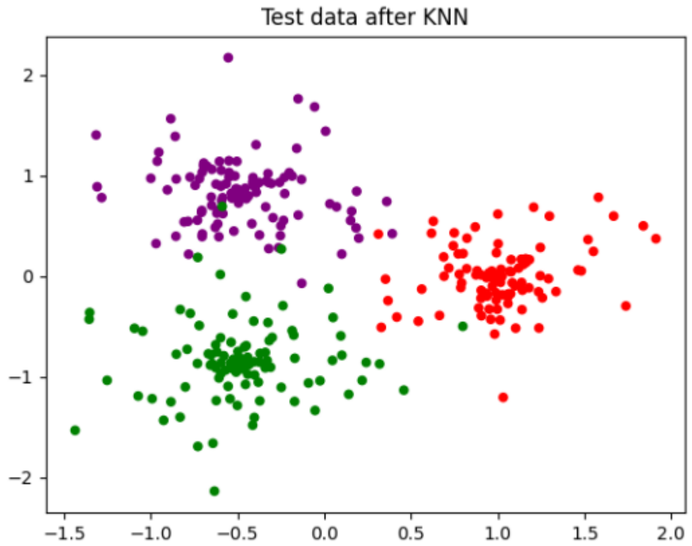
K-Nearest Neighbors

2. Example of classication using KNN:

KNN classification algorithm consists of assigning the new data point into the group that occurs most often among its K nearest neighbors.

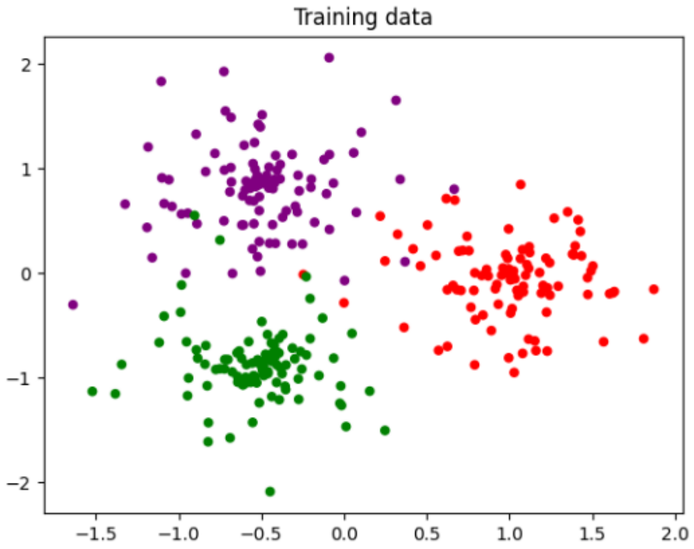


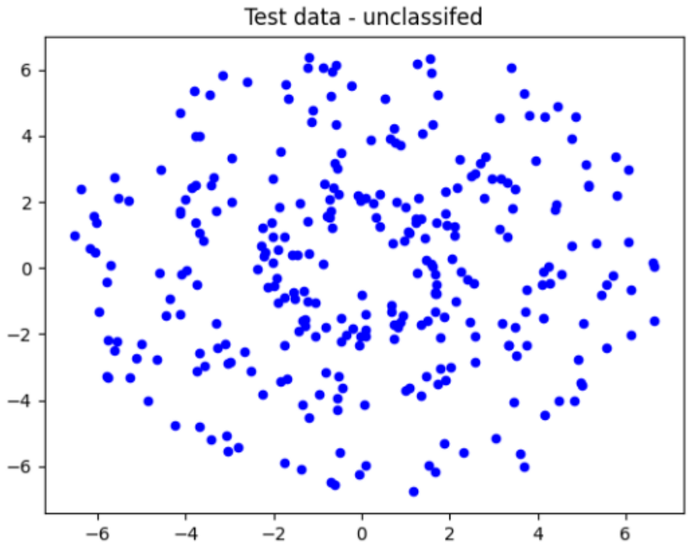


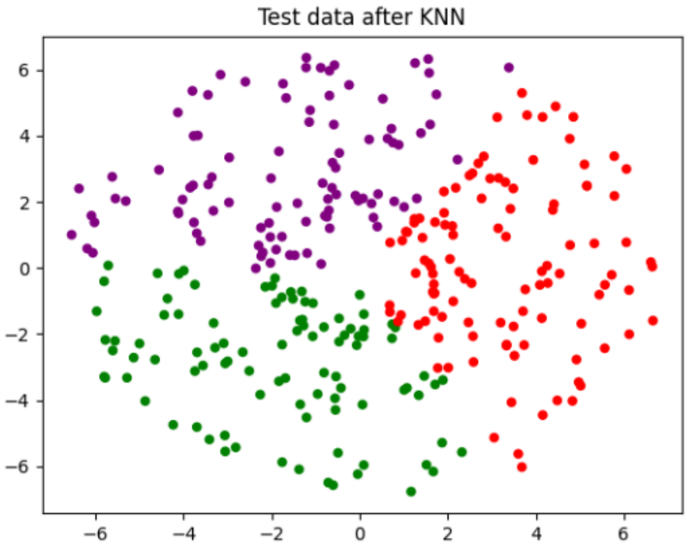


K-Nearest Neighbors

2. Example of classication using KNN:



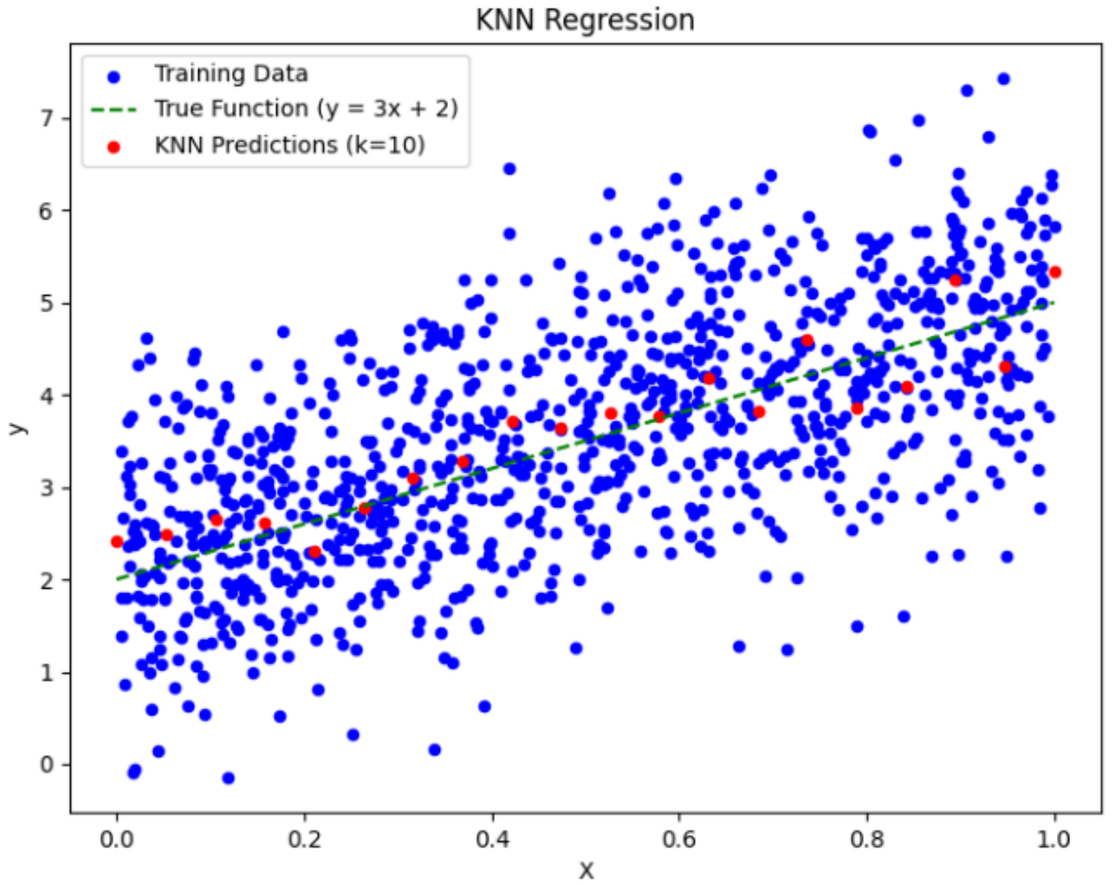




K-Nearest Neighbors

2. Example of regression using KNN:

The KNN (K-Nearest Neighbors) Regressor predicts values by averaging the target values of the nearest neighbors in the feature space.



KNN Clustering

END OF CHAPTER 2.2